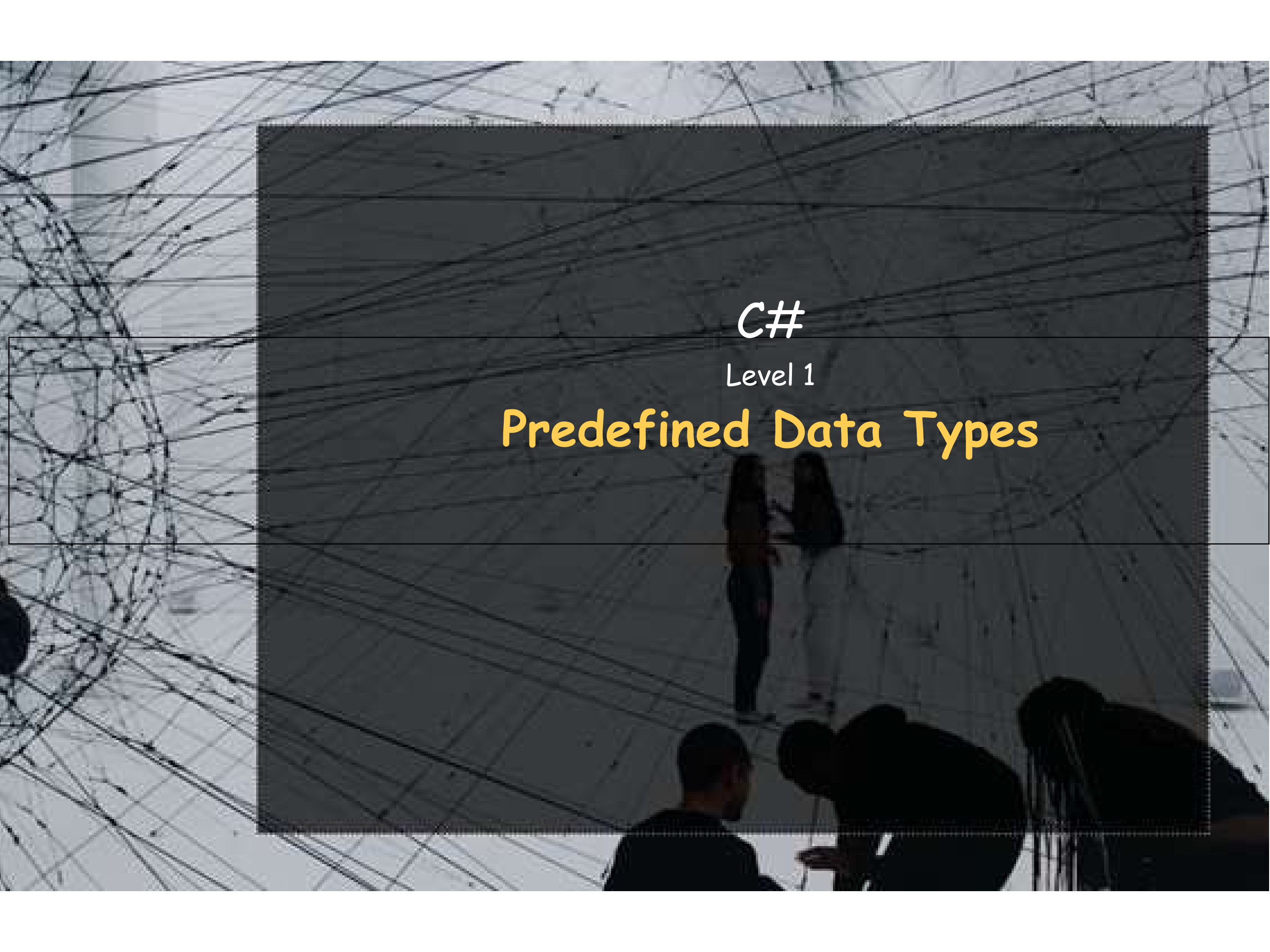


The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, white). At the bottom, a taskbar is visible with several application icons, including what looks like a web browser, a file explorer, and some development tools. The overall image is out of focus, serving as a backdrop for the text.

C#

Level 1

Predefined Data Types



C#

Level 1

Predefined Data Types

Definition

Built-In Data Types

-  Predefined data types are types already provided by *C#* and *.NET*.
-  They can be used directly without creating a custom data type.

Explanation

Main Predefined Categories

-  Integral Types → ``byte``, ``short``, ``int``, ``long``, ``sbyte``, ``ushort``, ``uint``, ``ulong``

-  Floating-Point Types → ``float``, ``double``

-  Decimal Type → ``decimal``

-  Character Type → ``char``

-  Boolean Type → ``bool``

-  String Type → ``string``

-  Object Type → ``object``

Integer Types

-  ``byte`` → small positive integers.

-  ``short`` → small integers.

-  ``int`` → commonly used whole numbers.

-  ``long`` → large whole numbers.

Procedures

1. Choose the predefined type that matches the required data.
2. Declare the variable using that type.
3. Assign a suitable value.
4. Use the variable in the program.

💡 Example

💡 Example

```
uint age = 25;
```

Note

- ◆ Integer types can be signed or unsigned.
 - ▣ Signed → `sbyte`, `short`, `int`, `long`
 - ▣ Unsigned → `byte`, `ushort`, `uint`, `ulong`
- ◆ Unsigned types cannot store negative values.

⊘ Warning

- ◆ Remember the required numeric suffixes:

float x = 10.5f;

long y = 1000000L;

decimal z = 10.5m;

- ◆ Do not confuse `object` with `var`:

object value = 10;

var number = 10;

▣ `object` is an actual predefined type.

▣ `var` is a keyword that lets the compiler determine the type.



Comparison

▨ Signed Integer Types

- ✓ Can store positive and negative values.
- ✓ Include ``sbyte``, ``short``, ``int``, and ``long``.

▤ Unsigned Integer Types

- ✓ Store zero and positive values only.
- ✓ Include ``byte``, ``ushort``, ``uint``, and ``ulong``.

✦ Summary

- ✓ Predefined data types are built into *C#* and .NET.
- ✓ Integer types include signed and unsigned types.
- ✓ ``float`` and ``double`` store floating-point numbers.
- ✓ ``decimal`` is used for high-precision decimal values.
- ✓ ``object`` is the base type for *C#* types.
- ✓ Numeric suffixes such as ``f``, ``L``, and ``m`` specify literal types.

Thank you 😊